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The impact of time devoted to video games on student achievement

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Abstract
The aim of this paper is to examine the relationship between the time that higher education students spend playing video games during exam periods and their average grades in one Eastern European country. Moreover, the authors wanted to explore the differences among students with regard to their age, gender, year of study, and employment status in relation to video game-playing habits. Four research questions were generated and a quantitative survey among students ($N=233$) was conducted at two universities in Serbia in December 2021. The results showed that, on average, students who play more games may have slightly lower grades than students who do not play games. In contrast, the time devoted to studying during the exam period is to some extent related to students' average achievement. In research sample, the best students spend the least time playing video games and the most time studying compared to other students.

Keywords Video games playing · Academic achievement · Average grades · Higher education · Eastern Europe

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1 Introduction

The world of video games is prone to frequent and rapid changes since it is strongly tied to the development of computer technologies, which, as we all know, are continuously evolving. Over the last two decades, video games have exploded in popularity. The use of video games, both by children and adults, has been identified as a noteworthy trend and a global success for video game firms. Between 2015 and 2020, the number of active gamers worldwide rose from 1.99 billion to 2.69 billion, and this figure is projected to continue increasing (Clement, 2021). The industry posted an estimated $180.3 billion revenue in 2021, and revenue is expected to grow at an average 8.7% over the next three years (The Games Market and Beyond in 2021, 2021). According to Entertainment Software Association, nearly 227 million Americans play video games. Across all ages, 80% of players are over 18, and the average age of a video game player is 31. Research results show that 51% of players spend seven plus hours playing games, 77% spend three plus hours and 29% spend between one hour and three hours playing per week on average (Entertainment Software Association, 2021). Similarly, in Europe 50% of population aged between 6 and 64 years old plays video games, equating to approximately 250 million players with 9.5 h average playing time per week. These stats demonstrate the relevance of video gaming in many people's daily lives. As a result, many scholars in the domains of computer science, psychology, sociology, education, and youth studies have conducted research on video games (Quwaider et al., 2019).

The increasing usage of video games and the amount of time young people devote to them has raised concerns that this kind of entertainment may begin to compete with academic time and, as a result, reduce academic achievement (Shin, 2004). A strong relationship between academic achievement and socioeconomic development and its great importance for wealth of nations and its prosperity (Steinmayr et al., 2014) initiated researchers' exploration of the effects of video game usage time on academic achievement (Adžić et al., 2021; Gnambs et al., 2020; Lei et al., 2022). Academic achievement refers to performance results that demonstrate how far a person has progressed toward specified goals that were the focus of activities in instructional settings, specifically in school, college, and university (Steinmayr et al., 2014). The majority of school systems specify cognitive goals that apply across several subject areas (critical thinking) or encompass the acquisition of information and comprehension in a specific intellectual domain (numeracy, literacy, science) (Steinmayr et al., 2014). Academic performance should be viewed as a multidimensional concept that encompasses various domains of learning. Because the subject of academic performance is so broad and encompasses such a wide range of educational outcomes, the definition of academic success is dependent on the indicators used to measure it. Although there are numerous indicators for measuring students' academic success, such as academic performance (grades, GPA, semester course, etc.), continuous academic performance (students' persistence, institutional retention, etc.), career success, academic advancement, attainment of learning outcomes, acquisition of skills, and competences persistence (Bloemer et al., 2017; York et al., 2015), the most frequently used is students GPA (Feldman & Kubota, 2015).

A GPA (Grade Point Average) is the average of all the grade points earned by a student throughout the course of his or her education and which can vary from 0.0 (the lowest) to 4.0 (the highest) (Feldman & Kubota, 2015). The evolution of research studies on the impact of video games on young adults has focused on studying the effects of video games on their behaviour and emotions, (Bowman & Tamborini, 2015; Granic et al., 2014; Nuyens et al., 2019; Przybylski & Weinstein, 2019; Zhang et al., 2021) as well as the effects of video games on academic achievement. (Adžić et al., 2021; Anand, 2007; Gnambs et al., 2020; Hawi et al., 2018; Lei et al., 2022; Nasution et al., 2015; Prena & Waver, 2020; Wakil et al., 2017). According to five dimension effect approach video games can effect players behaviour on at least five different dimensions – amount of play, content, context, structure and mechanics (Gentile, 2011). The amount of time play has been associated with lower academic performance (Weaver et al., 2013), game content has been shown to lead to a specific content learning (Gentile & Gentile, 2021), context of video game may moderate effects of other game characteristics on specific outcomes (Anderson et al., 2010; Anderson et al., 2019), game structure may affect videospatial processing (Castro-Alonso & Fiorella, 2019) and game mechanics can lead to improvements in specific motor skills (Merino-Campos & del Castillo Fernndez, 2016). This method has been proven to be beneficial in explaining why different research appear to find different types of impacts. According to the research findings, video game use has multiple beneficial and bad consequences on player behaviour (Greitemeyer, 2018; Kühn et al., 2019; Prescott et al., 2018; Przybylski & Weinstein, 2019; Zhang et al., 2021), as well as positive and negative effects on academic achievement (Adžić et al., 2021; Anand, 2007; Gnambs et al., 2020; Hawi et al., 2018; Lei et al., 2022; Nasution et al., 2015; Prena & Waver, 2020; Wakil et al., 2017).

Despite the fact that numerous studies have been conducted to investigate the relationship between video games and student academic achievement, the majority have been conducted in the context of Western countries; only a few are available in the context of Eastern Europe. Out of 16,700 scientific publications related to video games and student GPA, Google academic research database analysis revealed 3,533 scientific research publications on the relationship between student GPA and video games in twelve Eastern European countries. According to Jordan (2005), Eastern Europe consists of the four European countries of the former USSR and the eight countries of Southeast Europe (i.e. the Balkans). The number of studies per country was: Russia 1,920, Ukraine 1,030, Belarus 206, Moldova 150, Serbia 11, Bosnia and Herzegovina 5, North Macedonia 65, Montenegro 8, Albania 9, Bulgaria 16, Romania 37 and Greece 76. This gap is also recognized and analyzed within A Manifesto of European Video games (Haggis-Burridge et al., 2018). Considering that the European video game industry is rapidly and continuously evolving, the authors decided to conduct this research related to video games and student GPA in Serbia. Furthermore, by reviewing current research papers on the impact of video games on academic accomplishment, the authors discovered that the majority focused on high school and college students, with very little study undertaken with university students (Barr, 2020; Prena & Waver, 2020). Since amount of play is the most commonly used dimension of the five dimensions effect approach, and GPA is the most commonly used academic achievement factor, the authors decided

to investigate their link during the exam period among university students in the Republic of Serbia. At universities in Serbia, a six-point grading scale is used, where 10 (excellent-outstanding), 9 (excellent), 8 (very good), 7 (good), 6 (sufficient, satisfying—the lowest pass grade) and 5 (not sufficient, not satisfying –failure to grade). The required percentage is regulated by the statute of each individual institution, but general rules apply: 10 (91–100%), 9 (81–90%), 8 (71–80%), 7 (61–70%), 6 (51–60%), 5 0–50%). When calculating average grade, two decimals show up, e.g. 9.54 (Academic Grading in Serbia, 2022) Accordingly, the purpose of the study was to investigate the relationship between video game usage time during exam periods and students' GPA in higher education in Serbia. Furthermore, the authors intended to see if there were any causal factors between video game usage time and age, year of study, gender, and work status among university students during the exam period. A wide range of perspectives on the impact of video games on players, many scholars are interested in learning more about these themes in the scientific community. As a result, the demand for research, as well as the synthesis and revision of previously discovered findings, is increasing.

2 Literature review of the effects of video games on students' success

Based on the overview of the Google Academic data base for the period between 2000 and 2022 there are 16,700 scientific publications that are related to Video games and students GPA. In the last ten years that number has doubled; that is, between 2000 and 2010 there were 5,400 publications regarding this topic and between the 2010 and 2022 that number increased by twofold, up to 11.300 articles. To date, study findings on the influence of video games on academic achievement are mixed. While some studies demonstrate that video games have a detrimental influence on students' academic success, (Anand, 2007; Jackson et al., 2011; Weaver et al., 2013) others show that video games have a favourable impact on students' academic accomplishments (Kovess-Masfety et al., 2016; Posso, 2016). The majority of research studies on video games' effects were focused on examining their impact on young adults' behaviour and emotions. In this regard, the majority of first research studies on video games' effects were focused on examining the impact of violent video games on young adults' behaviour. Research findings showed that playing violent video games can influence aggression, depression, and antisocial behaviour. (Greitemeyer, 2018; Groves & Anderson, 2015; Krahé, 2014; Prescott et al., 2018; West et al., 2020; Zhang et al., 2021). However, recent research findings on impact of violent video games showed that there are no special changes in interpersonal competences, impulsivity, depressivity, anxiety, or executive functions behaviour among players (Ferguson et al., 2015; Kühn et al., 2019; Przybylski & Weinstein, 2019). The research findings that show that players who prefer violent video games consider them tool for reducing level of aggression and improving mood are particularly interesting (Bowman & Tamborini, 2015). Studies that were focused on examining the relationship between violent video games and students

GPA also showed inconclusive results (Burgess et al., 2012; Gentile et al., 2004). According to a research study of 607 eighth- and ninth-grade students from four different high schools, adolescents who spend more time playing violent video games are more likely to demonstrate anger, hatred toward instructors, and perform poorly in school (Gentile et al., 2004). Unlike previous research, a study on the influence of violent video games on students' GPA among 671 college students found no significant association (Burgess et al., 2012). Over the years the nature of video games has changed considerably, becoming increasingly complex, diverse, realistic, and social in nature (Ferguson & Olson, 2013), which has initiated a small but important body of research on the advantages and positive effects that video games can provide (Granic et al., 2014).

Research on general benefits of playing video games focuses on four primary areas: cognitive (attention), motivational (resilience in the face of failure), emotional (mood management), and social (prosocial behaviour) (Granic et al., 2014). So far, theoretical and empirical findings have outlined different paths as to how computer gaming might affect academic achievement. Time spent gaming might negatively affect academic performance via a time displacement mechanism, via attentional mechanisms, and/or via reduced sleep efficiency mechanisms (Drummond & Sauer, 2020). According to the time displacement mechanism, computer gaming replaces time that should be invested in academic activities (Hartanto et al., 2021; Subrahmanyam & Renukarya, 2015). Research results show that students who spend more time playing video games spend less time doing homework, completing assignments, and studying, which has a negative impact on their academic achievement (Hawi et al., 2018; Hellström et al., 2012; Sahin et al., 2016). However, research results also show that there is no significant relationship between time spent video gaming and academic achievement (Nasution et al., 2015; Tirgar et al., 2015). The research findings indicate that playing video games during the week has a detrimental impact on students' academic achievement, but playing video games at weekends has a beneficial impact. (Drummond & Sauer, 2020; Hartanto et al., 2018; Prena & Waver, 2020). Similar to the previous study, research into the amount of time students spend playing video games revealed that playing video games for one to three hours had no effect on GPA, while after playing video games for more than three hours GPA decreases, indicating a substantial difference between high and low video game consumption (Wakil et al., 2017). Research conducted in a small Saudi Arab public university showed that students' with high academic achievement (GPS) spend more time playing video games, almost as much as they spend on learning activities, and surprisingly they still earn high academic grades (Adžić et al., 2021). One of the most significant studies on impact of video game usage on academic achievement conducted among 219,000 students showed that there is no significant relationship between video game usage and academic achievement (Drummond & Sauer, 2020). These results confirmed a previous research study conducted among 192,000 students in 22 different countries (Drummond & Sauer, 2014). Previous research found just one significant difference between playing video games throughout the week and on weekends. Furthermore, research results showed that students who play video games before school on weekdays show a slight deterioration in

academic achievement, while students who played video games on weekdays after school show no changes in academic achievement.

According to the sleep displacement mechanism, computer gaming reduces quantity and quality of sleep (Altintas et al., 2019). Inadequate amounts of sleep contribute to poorer cognitive processing and attention deficit (Altintas et al., 2019; Wolfe et al., 2014), and may cause poorer academic performance. A study conducted among 2097 Lebanese high school students found that students who play video games and sleep five hours per day have a lower GPA than students who play video games but sleep seven hours per day on average (Hawi et al., 2018). According to the attention displacement mechanism, computer gaming may cause attention problems, lowering self-control and increasing impulsiveness (Gentile, 2011; Gentile et al., 2012). Several research studies have found that students who spend more time playing video games have attention deficit difficulties (Chan & Rabinowitz, 2006; Nikkelen et al., 2014). Unlike research studies that found detrimental impacts of video games, several studies have found favourable benefits of video games, particularly on cognitive development (Powers et al., 2013). According to developmental research, executive function develops from the age of seven through adolescence and is linked to scholastic success (Best et al., 2011).

Video games have the potential to develop certain skills because they are one of the tools for developing strategic and multidimensional thinking, problem solving, planning, teamwork abilities, and leadership traits (Silva & Mousavidin, 2015). A systematic review of 32 papers empirically investigating the cognitive benefits of video game playing showed that empirical research has underlined three specific skills over the years (Nuyens et al., 2019). These skills are: taskswitching (Dobrowolski et al., 2015; Pereg et al., 2013), attentional control (Cain et al., 2014; Mack et al., 2016; Schubert et al., 2015), and subsecond time perception (Donohue et al., 2010; Rivero et al., 2013). It has been suggested that video gaming can improve cognitive processes (e.g., improved attention control and processing speed) (Nuyens et al., 2019) and postpone cognitive decline (Griffiths et al., 2013). Research conducted over the course of four high school years in a sample of 1,492 teenagers showed that more strategic video game play predicted stronger self-reported problem-solving skills. Furthermore, the findings supported an indirect relationship between strategic video game play and higher academic grades, (Adachi & Willoughby, 2013). One of the limitations of present research studies is the preponderance of cross-sectional research designs, which limits effective data analysis and interpretation (Gnambs et al., 2020), and which indicates that the implementation of longitudinal research studies would be more appropriate in examining the effects that video games have on students' academic achievement. One of the few longitudinal research studies, which lasted three years and was conducted in 176 schools in Germany among 3554 students, found that playing video games on a daily basis had no significant impact on students' academic achievement (Gnambs et al., 2020). Regardless of the increasing number of research studies on effects that video games have on academic achievement, results remain inconclusive.

3 Methodology and sample

The objectives of this study are (1) to examine the impact of playing video games on students' academic performance and (2) to examine the gaming habits of the best students. In order to assess the opinions and attitudes of the respondents, primary research was conducted at two universities in Serbia, one of which is private and the other state-owned, during the period from December 6, 2021, to December 20, 2021. Data were collected by the survey method. A convenient sampling technique was applied. The online questionnaire "Students and Video Games" was filled out by 266 respondents, of which 233 were valid. Those respondents who have not taken the exams so far, as well as those who are not students because they did not state the year of study, have been removed from the database. The survey was anonymous, and participation in the student survey was on a voluntary basis.

The questionnaire consists of two parts. The first part consists of questions about the average time students spent playing video games during the day in minutes during the exam period, the average time respondents spent studying during the day in minutes during the exam period, the average student achievement (GPA), and the average time respondents spent during days in minutes at other types of entertainment. In the second part, there are questions of a general type: about gender, age, year of study, whether they are employed or otherwise. From the point of view of measurement, the variables are Nominal, Ordinal, and Scale.

233 respondents participated in the primary research. Of these, 158 play video games, while 72 do not. Observed by age of students, 67.4% are 19–22 years old, 20.2% 23–26 years old, 6.0% 27–30 years old, and 6.4% are older than 30 years old. The largest number of students, over 40% in the sample, are in the second year of their undergraduate studies, while the smallest number of students are undertaking doctoral studies. In terms of employment status, 72.1% of respondents are not employed. There are almost equally male and female respondents in the sample (for a more detailed overview of demographics, see Table 1).

This study raised the following research questions:

Q1. Does the time spent playing video games affect student success?
Q2. Does the time spent studying affect student success?
Q3. Are the students with the highest grades the biggest gamers?
Q4. What is the relationship between playing video games and other characteristics of respondents: year of study, employment status, age, and gender?

To obtain answers to the research questions, the following statistical techniques were used: Chi-squared test for independence, Independent samples t-test, One-way analysis of variance (ANOVA), Correlation (Pearson Product-Moment Correlation Coefficient), and Regression (Bivariate Linear Regression). The data were processed using the IBM's SPSS Statistics v. 25 for Windows software package.

Table 1 Summary of the demographics of the sample

		Frequency	Percent
Do you play games?	1 Yes	158	67,8
	2 No	75	32,2
	Total	233	100,0
What is your age?	19–22	157	67,4
	23–26	47	20,2
	27–30	14	6,0
	31 +	15	6,4
	Total	233	100,0
What year are you studying?	1 1st	7	3,0
	2 2nd	97	41,6
	3 3rd	54	23,2
	4 4th	43	18,5
	5 Partial fulfilment	18	7,7
	6 Master student	10	4,3
	7 PhD student	4	1,7
	Total	233	100,0
Are you employed?	1 Yes	60	25,8
	2 No	168	72,1
	3 Other	5	2,1
	Total	233	100,0
What is your gender?	1 Male	113	48,5
	2 Female	116	49,8
	3 Other	4	1,7
	Total	233	100,0

4 Data analysis and research results

The average age of students is 22.9 years. The youngest student is 19 years old, and the oldest is 54 years old. The average grade of the respondents is 8.51, and ranges from 6.71 to a maximum of 10. There are students who admitted that they did not spend a minute preparing for their exams, and there are also those who studied 15 h a day. On average, they studied for 258 min, or 4 h and 18 min. They spent a little less than 4 h on the fun activities, averaging 224 min. The maximum number of minutes spent in video games, going out, hanging out with friends, etc. was 840 min, while the maximum time that student gamers spent only on playing video games (approximately one-third of the sample) was exactly 9 h. Of course, there were students who did not spent time on fun or video games. The average playing time among students who played video games during the exam period was just over an hour (72 min to be exact).

4.1 Q1: Does the time spent playing video games affect student success?

By observing the normal Q-Q plot, we tested the normality of the distribution of the dependent variable (GPA). A reasonably straight line suggested a normal distribution. An independent samples t-test was conducted to compare the average grades of students who play games ($n = 158$) and students who do not play games ($n = 75$). The test was significant, $t(227) = -3.285, p = .001$. Levene's test for equality of variances was not significant ($p > 0.05$), indicating equal variances. Student gamers from our sample ($M = 8.39, SD = 0.874$) recorded lower success during their studies than their non-gaming counterparts ($M = 8.78, SD = 0.680$). The magnitude of the differences in the means ($meandifference = -0.38, 95\%CI : -0.61do - 0.15$) was very small ($Cohen^{'}sd = .18$). The results indicated that, on average, students who play more games had lower grades than students who do not play games. However, the size of the difference is very small, it is just one of the factors that could affect students' grades. From Table 2 it can be seen that the average learning time of the first group of students (gamers) is lower than the average learning time of the second group of students (non-gamers).

The relationship between current GPA and time spent in games was investigated using the Pearson product-moment correlation coefficient. There was a weak, negative correlation between these two variables, but it was not significant ($r = -.122, n = 229, p = .065$). There was also a statistically insignificant correlation between the current GPA and the time spent on all fun activities, $r = -.106, n = 228, p = .110$. We calculated the variable of fun activities by summing up two variables from the survey, the average time spent playing video games during the day and the average time spent on other types of entertainment during the day. Given that the correlations were not statistically significant, the conditions for conducting linear regression tests were not met.

4.2 Q2: Does the time spent studying affect student success?

Using the Pearson linear correlation coefficient, the relationship between students' GPA and the average time students spent learning was investigated. GPA was significantly related to the time students spent studying, $r = .335, n = 228, p < .001$. The direction of the relationship between the variables is positive, the strength of the relationship is medium, and the significance level is high. The question is whether it is possible to predict how much time one should study for, which is a dependent

Table 2 Average gaming, fun, and studying time for gamers and non-gamers

		What is your current GPA?	Average gaming in minutes/day	Average fun in minutes/day	Average studying in minutes/day
		Mean	Mean	Mean	Mean
Do you play games?	1 Yes	8.39	101.77	246.20	235.64
	2 No	8.78	9.80	178.20	305.60

variable, for certain values of the independent variable, i.e., GPA scores? Correlation coefficients do not indicate the direction of causality. Therefore, the next step in the analysis should be regression analysis.

The first step is to create a scatterplot to assess whether the basic assumptions for simple linear regression are met. The relationship between the variables appears to be linear, which ensures that there is no violation of the linearity assumption. The points on the diagram are grouped around a straight line. As the value of one variable increases, so do the values of the other variable, ensuring that there is no violation of the assumption of homoscedasticity. Since it was determined that the relationship between the variables is approximately linear and that the results are evenly distributed, we can proceed with regression analysis. The results of the analysis suggest that it is possible to predict what the GPA will be based on the average daily time devoted to learning during the exam period. The regression equation for predicting student's GPA is $PredictedGPA = 8.074 + 0.002LearningTime$. It should be noted that the 95% confidence interval for the slope is fairly narrow $(0.001 - 0.002)$ and does not contain the value of zero ($p < .001$), which suggests that the time devoted to learning during the exam period is to some extent related to average student achievement. The constant of 8.07 for zero time studying suggests that time studying is, at best, a weakly correlating variable. Accuracy in predicting average student achievement is moderate. The correlation coefficient R is 0.335, which means that R squared is. 112. Approximately only 11% of the variance of the predicted GPA was accounted for by its linear relationship with the learning time.

We interpret the regression equation as follows. The value of the constant in the regression equation of 8.07 represents the y-intercept of the line and can be interpreted as GPA when the value of the independent variable is 0, i.e., the average GPA without learning in the exam period. The slope (or gradient) of the line, 0.002, represents the expected change in the value of the dependent variable (outcome) resulting from the unit change in the value of the independent variable (predictor).

4.3 Q3: Are the students with the highest grades the biggest gamers?

To gain an answer to this research question, the first step is to divide students from our sample into three groups according to the grade point average: weak students with an average grade of 6 to 7.49 ($n = 28$), average students with an average grade of 7.50 to 8.99 ($n = 121$), and the best students with an average grade of 9 to 10 ($n = 84$). From Table 3 it can be seen that, in researched sample, the best students spend the least time playing video games, and spend the most time learning compared to other groups of students, regardless of whether they are gamers or otherwise. A chi-squared test for independence indicated a significant association between students' GPA (with three levels) and video game-playing status (with two levels) but with a very low impact, $\chi^2(2, n = 223) = 11.54, p = .003, Cramer's V = .034$. As expected, a one-way analysis of variance indicated a statistically significant difference between the three groups of students (weak, average, and best) in relation to the current average grade of students, $F(2, 226) = 628.997, p < .001$. The strength of relationship between these groups was quite strong, since η^2 equals 0.848 (Table 3).

Table 3 Differences between students grouped by success in relation to their gaming habits

			What is your current GPA?	Average gaming in minutes/day	Average fun in minutes/day	Average studying in minutes/day
Do you play games (PC, PS, XBOX, phone)?	Yes	Weak s. (6.00–7.49)	7.08	96.31	227.27	165.38
		Average s. (7.50–8.99)	8.18	108.43	268.17	233.78
		Best s. (9.00–10.00)	9.45	93.37	219.49	276.02
	No	Weak s. (6.00–7.49)	7.22	0.00	315.00	150.00
		Average s. (7.50–8.99)	8.28	13.82	170.92	251.05
		Best s. (9.00–10.00)	9.36	6.00	178.29	373.71
	Subtotal	Weak s. (6.00–7.49)	7.09	89.43	233.54	164.29
		Average s. (7.50–8.99)	8.21	78.72	237.38	239.25
		Best s. (9.00–10.00)	9.41	56.96	202.32	316.73

4.4 Q4: What is the relationship between playing video games and other characteristics of respondents: year of study, employment status, age, and gender?

A chi-squared test for independence was conducted to assess the relationship between video game-playing status and the year of study by category $(2 \times 7 contingency table)$, $\chi^2(6, n = 223) = 11.23, p = .081, Cramer's V = .22$. The two variables were not significantly related. There was also no significant relationship between playing video games and employment $(2 \times 3 contingency table)$, $\chi^2(2, n = 223) = 2.96, p = .228, Cramer's V = .11$, as well as between playing video games (with two levels) and age categories of respondents (with four levels), $\chi^2(3, n = 223) = 5.53, p = .137, Cramer's V = .15$.

However, a $2 \times 2 contingency table$ analysis (with Yates' Continuity Correction) showed a significant relationship between video game-playing status and the gender of the respondents (four respondents who declared themselves as "Other" were removed from the database due to testing conditions) with a medium impact, $\chi^2(1, n = 229) = 51.61, p < .001, Phi = .48$. In our sample, nine out of ten male students certainly played games, while less than half of female students practiced the same (Fig. 1).

The logical next step was to conduct a series of t-tests to determine whether there is a significant statistical difference in gender in relation to GPA, in relation to average learning time, as well as in relation to average playing time (Table 4). The mean difference of all tests was significant at the 0.001 level. On average, the female population

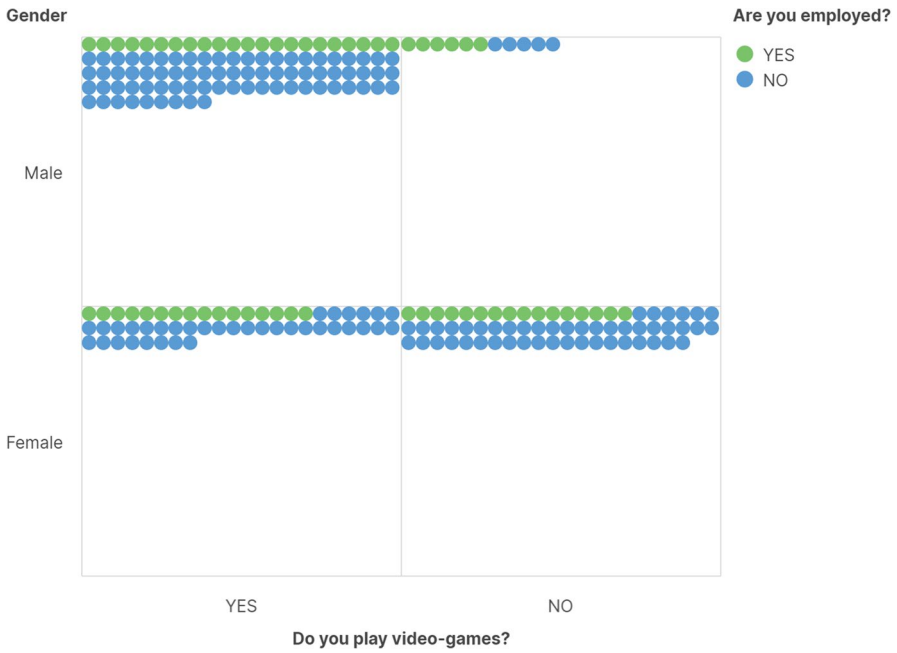

Fig. 1 Number of male and female students in relation to whether they are gamers and employed

Table 4 Studying, playing games, and GPA in relation to the gender of the respondents

		Do you play games (PC, PS, XBOX, Nintendo, phone)?		What is your current GPA?	Average studying in minutes/day	Average gaming in minutes/day
		1 Yes	2 No	Mean	Mean	Mean
		Count	Count			
What is your gender?	1 Male	102	11	8.33	223.75	111.59
	2 Female	52	64	8.72	294.01	31.12

in the sample ($M = 8.72, SD = 0.778$) achieved a higher average GPA than the male population ($M = 8.33, SD = 0.846$), $t(223) = -3.620, p < .001, Cohen' sd = .18$. One of the possible reasons for higher grades of female students compared to male students could be found in the analysis of time spent in learning and time spent in video games of our sample. In relation to learning time, male students ($M = 223.75, SD = 150.685$) spent 25% less time learning than female students ($M = 294.01, SD = 169.091$), $t(226) = -3.308, p = .001, Cohen' sd = .18$. In relation to gaming time, male students ($M = 111.59, SD = 107.191$) spent almost four times longer playing video games than female students ($M = 31.12, SD = 50.901$), $t(159.13) = 7.226, p < .001, Cohen' sd = .26$. Levene's test indicated unequal variances, so degrees of freedom were adjusted from 227 to 159.13. Not only were men bigger players than women, but younger men and younger women were bigger players than older ones. Namely, the biggest players in the sample were first-year students, both men ($M = 210.00, SD = 42.426$) and women ($M = 46.00, SD = 63.087$). Year of study appears to be a corrective factor as fourth-year male students ($M = 75.79, SD = 75.742$) and third-year female students ($M = 14.55, SD = 33.433$) spent the least amount of time playing video games. Although the differences between the groups were statistically significant, the effect size between the groups was not large, because many other factors can affect the difference in academic achievement of male and female students, not just time spent learning and playing video games.

5 Discussion

The first part of the questionnaire provided us with an intriguing profile of the university population "gamer." Despite the fact that various studies have been undertaken in order to determine a gamer profile, (Bateman & Boon, 2005; Grooten & Kowert, 2015; Kirman & Lawson, 2009; Manero et al., 2016; Mulligan & Patrovsky, 2003; Paaßen et al., 2017; Steltenpohl et al., 2021) the authors of this study have decided to define a gamer as a student who played video games during the exam period, regardless of the amount of time spent playing games. The study findings related to first research question show that there are significantly more students

who play games during the exam period than those who do not. The difference between these two groups of students was statistically proven at the 0.001 level. On average, the 158 students who play games had lower grades than the 75 students who do not play games. One of the possible reasons for the lower success rate of the gamer student group is the fact that they spent less time learning, averaging 236 min a day, compared to non-gamers, who studied an average of 306 min a day during the exam period. This finding is consistent with the time displacement mechanism, which states that students who spend more time playing video games spend less time doing homework, completing assignments, and studying, resulting in lower academic achievement (Hartanto et al., 2021; Hawi et al., 2018; Sahin et al., 2016). Thus, the first group of gamer students achieved a lower average grade of 8.39 compared to the second group of non-gamer students, whose average grade was 8.78. However, although statistically confirmed, the discrepancy in GPA is neither large nor important. Moreover, since the magnitude of the mean differences between the two variables (average game time and average grade) was small, and the correlation between the two variables was weak and not significant, it is easy to assume that the time of playing video games is just one of the factors that can influence the academic achievements of students in our research sample. These findings are consistent with previous research results that demonstrated a relationship between video game usage and lower GPA. Only the outcomes of that study were significantly more severe among elementary and secondary school students (Drummond & Sauer, 2014, 2020; Gentile, 2011) as than among university students (Adžić et al., 2021; Barr, 2020; Prena & Waver, 2020). It suggests that any detrimental effect that video games may have on academic achievement among elementary and secondary school students is related to the development of physical and cognitive abilities. (Domingues-Montanari, 2017; Hale & Guan, 2015). Lower academic performance among university students, on the other hand, might be the result of individual differences (Steinmayr et al., 2014) or poor time management skills (Ahmad et al., 2020; Alani et al., 2020; Indreica et al., 2011; Nasrullah & Khan, 2015; Razali et al., 2018; Weaver et al., 2013). Nevertheless, the association between video game playing and academic success might be attributed to an unknown reason or to personal qualities of each individual player such as IQ, psychological state, social status, or motivation (Steinmayr et al., 2014).

The second study question aimed to find out whether the time students devote to learning results in higher grades. The direction of the relationship between the variables is positive, the strength of the relationship is medium, and the level of significance is high. The regression equation at the same significance level of 0.0005 suggests that for every minute of learning a student in our sample could expect an increase in the average grade by 0.002. A constant value of 8.074 represents the average grade at the beginning of the exam period. A result of 8.07 for zero time studying, on the other hand, implies that time studying is, at most, a weakly correlated variable. The study showed that, there was weak correlation between study time and higher GPA. This is not surprising, as it might be explained by a variety of previously researched variables such as demographic status, IQ, behavioral features, and psychological factors such as attitudes, self-esteem, self-efficacy, and self-concept. (Steinmayr et al., 2014). In addition to these difficult-to-control factors, each

student has a unique learning style that might affect their academic success. (Al-Zoubi & Younes, 2015).

Third research question was aimed to examine the gaming habits of the best students. In order to get better insight the students were categorized into three groups according to the average grade: 28 weak students with an average grade from 6 to 7.49, 121 average students with an average grade from 7.50 to 8.99, and 84 best students with an average grade from 9 to 10. The results of the analysis of the third research question showed that the best students spend the least time playing video games, and the most learning compared to other groups of students, regardless of whether they are players or not. But just comparing the best who didn't play games to the best who did, those who did gained a higher average grade (according to Table 3), and indeed spent less time studying. On average, the best students achieved an average grade of 9.41 with an average playing time of only 57 min, compared to 317 min of learning per day. In order to achieve an average grade of 8.21, average students played more games, 79 min per day, and studied less, 239 min per day. Finally, the weakest group of students in our sample achieved an average grade of 7.09. These students devoted the least time to learning, averaging 164 min per day during the exam period, compared to the most time devoted to playing video games, 89 min on average per day. Therefore, the results indicated that the weakest students are the strongest players and not the other way around. But are they playing more games because they are just weaker students, or weaker students because they are playing more games? The new findings support prior research results that excessive video gaming is associated with decreased academic achievement (Gnambs et al., 2020; Hartanto et al., 2018). Furthermore, the study findings revealed that playing video games for only an hour and twenty-nine minutes per day during the exam period had a negative impact on academic accomplishment, in contrast to the prior study (Wakil et al., 2017) findings that indicated that playing video games for one hour to three hours per day has no effect on GPA. It would be interesting to examine whether there is a difference between playing video games before and after courses throughout the week, given recent study by Drummond and Sawer revealed a minor decline in academic success among students who played before classes (Drummond & Sauer, 2014).

The abundance of collected data enabled us to define another research question, Q4, i.e., to examine the relationship between playing video games and certain demographic characteristics of respondents, such as year of study, employment status, age, and gender. The only relevant statistical difference was found in the relationship between the status of playing video games and the gender of the respondents. The probable reason for this difference in our sample is the fact that nine out of ten male students played games compared to five out of ten female students. The male students spent almost four times longer playing video games than the female students. According to the ESA (Entertainment Software Association, 2021), the average age of video game players in the United States is 31 years old. However, our results show that if we restrict the analysis to the university students population the most numerous players were the 19 years old freshmen. Furthermore, first-year students' average play time of 3.5 h is higher than the average play time of 3 h per player from the ESA study. It seems that the year of study acts as a corrective factor because

students of higher years spend less time playing games. Moreover, the male students spent 25% less time learning than the female students. Thus, on average, with a significance level of 0.001, the female population of 116 in the sample achieved a higher average grade than the male population of 113 in the sample, or 8.72 and 8.33, respectively. Gender differences with regard to video game usage time have been confirmed in numerous studies (Leonhardt & Overå, 2021; Veltri et al., 2014). Overall, the literature shows that male gamers start playing video games earlier in life, play more frequently, and spend more time playing video games than female gamers (Borgonovi, 2016; Leonhardt & Overå, 2021). Only a few research on female gamers in Europe have been published. (Griffiths & Lewis, 2011; McLean & Griffiths, 2013; Shaw, 2012). According to the findings of these qualitative research, women often play casual games for shorter durations of time than males, yet half of these studies also say that female gamers play MMORPGs (Griffiths & Lewis, 2011). Furthermore, gaming motivations vary across genders and cultures (Griffiths & Lewis, 2011; Ko et al., 2005; Laconi et al., 2015; Taylor, 2006). Overall, females and males have distinct gaming motives. (Lopez-Fernandez et al., 2019). According to Entertainment Software Association 52% of players identify themselves as male and 48% of players identify themselves as female (Entertainment Software Association ESA, 2021). As the number of female video gamers grows, it is critical to comprehend their motivations and behavior (Leonhardt & Overå, 2021).

The aim of the current study was to examine the association between time spent video gaming and students' GPA during their exam periods among university students in the Republic of Serbia. Furthermore, the aim of the study was to examine the relationships between video game usage time and age, year of study, gender, and work status among university students. The presented results offer intriguing possibilities for future research. First, the study focused on a specific, most commonly used academic achievement measure (GPA) and the time spent on video games during the exam period. Aside from the quantitative approach, it appears necessary to pay closer attention to qualitative aspects of video games such as game content, game context, game structure, and game mechanics. It appears worthwhile to investigate certain other possible aspects of video games in order to establish the extent to which they influence students' academic progress. Even though our study did not include a longitudinal view on gaming periods, future research should use a balanced longitudinal design to investigate how changes in video gaming are related to changes in academic accomplishment. Furthermore, as in other studies, we relied on self-reported gaming times. However, the validity of these metrics has lately been called into doubt since many computer gamers underreport the amount of time they spend playing computer games (Kahn et al., 2014). As a result, it may be beneficial in the future to investigate the possibilities provided by behavioural data such as procedural data (e.g., computer-produced log files) that may provide further insights into the characteristics of students' computer gaming time. Finally, while we included reasoning ability as a predictor of academic achievement, it may be worthwhile to include additional individual variations that may reduce the potentially negative consequences of computer gaming. As a result, future research should incorporate appropriate measures of individual variations. (e.g., self-regulation, academic motivation) as moderators in their gaming research.

6 Conclusion

The current study findings contribute to the limited empirical evidence on the relationship between video gaming and university students' GPA. The research showed video game playing to be associated with lower academic achievement to some degree. Consistent with most previous research, the present study findings lead us to understand how time spent video gaming effects students GPA among university students during the exam period. Since the most of research studies have been undertaken in Western countries with elementary and secondary school students, the researchers believe that the findings of this study will shed light on the influence of video game usage time on academic accomplishment in Eastern European universities. In conclusion, there was a distinct and significant difference in the reported GPAs between high usage video game players and low usage video game players as well as gender differences. However, in addition to the amount of time spent playing video games, a variety of factors such as past knowledge, skills, and abilities, as well as learning activities throughout the semester year, should be considered. Overall, it appears that the detrimental effects of computer gaming on academic achievement are minimal. It is still unclear whether casual computer gaming has any practical consequences for the majority of students. Rather, the public debate over the dangers of computer gaming appears to be exaggerated.

It is very important to carefully consider the limitations of the study and we believe that there are several. The first limitation is certainly that this study covers only one country with limited sample size. Further study on the impact of video game time on academic accomplishment should be focused on a larger sample size from the entire region in order the improve the limited generalizability of our results. Although GPA is the most commonly used indicator of student achievement, the second limitation refers to the study focusing on examining solely GPA as an indicator of student accomplishment. Since, there are numerous indicators for measuring students' academic success, such as academic performance (grades, GPA, semester course, etc.), continuous academic performance (students' persistence, institutional retention, etc.), career success, academic advancement, attainment of learning outcomes, acquisition of skills, and competencies persistence (Bloemer et al., 2017; York et al., 2015), the most frequently used is students GPA (Feldman & Kubota, 2015) further research studies should focus on examining all these indicators to obtain better insight. The third limitation refers to the study focusing on only one dimension effect of video games on players' behavior. Amount of play (time spent playing video games) is certainly the most commonly used dimension and has been associated with lower academic performance but further research study should encompass at least five different dimensions amount of play, content, context, structure, and mechanics (Gentile, 2011). This method has been proven to be beneficial in explaining why different research appears to find different types of impacts. Furthermore, we relied on self-reported playing times, as in prior studies. However, the accuracy of these assessments has lately been called into doubt since many computer gamers underreport the amount of time they spend playing computer games (Kahn et al., 2014). The fourth limitation refers to the research instrument. It appears

worthwhile to investigate the reasons for certain behavior to establish the extent to which they influence students' academic progress. The research instrument should be standardized and incorporate text-based questions to obtain the best possible results. Despite the fact that the study findings during the exam period were effective, this may be considered as one of the limitations. Students' stress levels increase when they enroll at university and do not return to pre-university levels during their stay (Bewick et al., 2010). University students were found to have greater rates of psychological distress than the general population (Eskin et al., 2016). Furthermore, stress is the most frequently reported constraint to students' academic progress. (ACHA, 2019) Since the majority of video game players (89%) play to relieve stress (Entertainment Software Association ESA, 2021), future study should include the whole school year to provide a comprehensive analysis. Finally, limitation certainly refers to the implementation of a cross-sectional research design, but future research should use a balanced longitudinal design to investigate how changes in video gaming are related to changes in academic accomplishment.

Appendix: Questionnaire

1. Do you play video games on electronic devices (PlayStation, XBOX, mobile phone, PC, etc.)?

   Yes
   No

2. Have you taken exams in any exam period so far?

   Yes
   No

3. What is your current grade point average?

   * (open question)

4. How much time did you study PER DAY on average during the last exam period? How many hours and minutes?

   * (open question)

5. On average, how much time did you play video games PER DAY during the last exam period? How many hours and minutes?

   * (open question)

6. On average, how much time did you spend PER DAY on fun activities, going out, etc. during the last exam period, NOT INCLUDING the time you spent playing games? How many hours and minutes?

   * (open question)

7. What year are you studying?

   1st
   2nd
   3rd
   4th
   Partial fulfillment
   Master student
   PhD student
   Other

8. How old are you?

   * (open question)

9. Are you employed?

   Yes
   No
   Other

10. What is your gender?

    Male
    Female
    Other

Declarations

The authors have no relevant financial or non-financial interests to disclose.
The authors have no competing interests to declare that are relevant to the content of this article.
All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.
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